

Calculus II: Functions of Multiple Variables, the Gradient, and the Hessian

(Lecture Notes)

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Functions of Multiple Variables

So far, we have focused on functions with a single input, like $y = f(x)$. However, most real-world systems and machine learning models depend on many variables:

$$y = f(x_0, x_1, x_2, \dots, x_n)$$

A function with two inputs, like $z = f(x, y)$, can be visualized as a surface in three-dimensional space.

We can also have systems with multiple outputs, forming a **system of equations**:

$$\begin{aligned}y_0 &= f_0(x_0, x_1, \dots, x_n) \\y_1 &= f_1(x_0, x_1, \dots, x_n) \\&\vdots \\y_m &= f_m(x_0, x_1, \dots, x_n)\end{aligned}$$

Vector and Matrix Notation

Writing out large systems of equations is tedious. If the system is **linear**, we can use matrices and vectors to represent it compactly. A system like:

$$\begin{aligned}y_0 &= \theta_{00}x_0 + \theta_{01}x_1 + \dots + \theta_{0n}x_n \\y_1 &= \theta_{10}x_0 + \theta_{11}x_1 + \dots + \theta_{1n}x_n \\&\vdots \\y_m &= \theta_{m0}x_0 + \theta_{m1}x_1 + \dots + \theta_{mn}x_n\end{aligned}$$

can be simplified to:

$$\vec{y} = M\vec{x}$$

Here, $\vec{x} \in \mathbb{R}^n$ is the input vector, $M \in \mathbb{R}^{m \times n}$ is the matrix of coefficients (θ_{ij}), and $\vec{y} \in \mathbb{R}^m$ is the output vector. This represents a linear transformation from an n -dimensional space to an m -dimensional space.

Note on Neural Networks: A fully-connected layer in a neural network is essentially this: a linear transformation followed by a non-linear activation function, $\vec{y} = f(M\vec{x})$.

Derivatives of Multi-variable Functions

Partial Derivatives

For a single-variable function, the derivative represents the slope at a point. But for a multi-variable function (a surface), which direction's slope do we measure? The simplest approach is to find the rate of change along one of the axes while holding the other variables constant. This is called a **partial derivative**.

For a function $y = f(x_0, x_1)$, the partial derivative with respect to x_0 is written as $\frac{\partial y}{\partial x_0}$. It's defined as:

$$\frac{\partial y}{\partial x_0} = \lim_{\Delta x_0 \rightarrow 0} \frac{f(x_0 + \Delta x_0, x_1) - f(x_0, x_1)}{\Delta x_0}$$

Example: Calculating Partial Derivatives

Let's use the function $y = 3x_0 + 4x_1^2 + x_0x_1$.

Method 1: Using the Limit Definition. Let's calculate $\frac{\partial y}{\partial x_0}$ from first principles.

$$\begin{aligned} \frac{\partial y}{\partial x_0} &= \lim_{\Delta x_0 \rightarrow 0} \frac{[3(x_0 + \Delta x_0) + 4x_1^2 + (x_0 + \Delta x_0)x_1] - [3x_0 + 4x_1^2 + x_0x_1]}{\Delta x_0} \\ &= \lim_{\Delta x_0 \rightarrow 0} \frac{3x_0 + 3\Delta x_0 + 4x_1^2 + x_0x_1 + (\Delta x_0)x_1 - 3x_0 - 4x_1^2 - x_0x_1}{\Delta x_0} \\ &= \lim_{\Delta x_0 \rightarrow 0} \frac{3\Delta x_0 + (\Delta x_0)x_1}{\Delta x_0} \\ &= \lim_{\Delta x_0 \rightarrow 0} (3 + x_1) \\ &= 3 + x_1 \end{aligned}$$

Method 2: Treating Other Variables as Constants. This is the practical shortcut. To find $\frac{\partial y}{\partial x_0}$, we treat x_1 as a constant and differentiate normally.

$$\frac{\partial y}{\partial x_0} = \frac{\partial}{\partial x_0}(3x_0 + 4x_1^2 + x_0x_1) = 3 + 0 + x_1 = 3 + x_1$$

As you can see, both methods yield the same result. The shortcut is a direct consequence of the limit definition.

Mixed Partial Derivatives

We can also take higher-order partial derivatives. This includes taking a derivative with respect to one variable, and then another. For our example, $\frac{\partial^2 y}{\partial x_1 \partial x_0} = \frac{\partial}{\partial x_1}(3 + x_1) = 1$. For continuous functions, the order of differentiation does not matter, so $\frac{\partial^2 y}{\partial x_1 \partial x_0} = \frac{\partial^2 y}{\partial x_0 \partial x_1}$.

Taylor's Formula for Multiple Variables

Deriving the Maclaurin Series for Two Variables

How can we approximate a multi-variable function with a polynomial? Let's take the Maclaurin series case (approximating around the origin) for a function $f(x, y)$. We can propose that our function can be approximated by a polynomial containing all combinations of the powers of x and y :

$$P(x, y) = b_0 + b_1x + b_2y + b_3x^2 + b_4y^2 + b_5xy + \dots$$

To make this a good approximation, we must match the value and the partial derivatives of $P(x, y)$ to our function $f(x, y)$ at the point $(0, 0)$.

1. **Match the function value:** At $(0, 0)$, all terms with x or y in $P(x, y)$ go to zero.

$$f(0, 0) = P(0, 0) = b_0 \implies \mathbf{b}_0 = \mathbf{f}(\mathbf{0}, \mathbf{0})$$

2. **Match the first partial derivatives:** First, we find the partial derivatives of our polynomial $P(x, y)$:

$$\frac{\partial P}{\partial x} = b_1 + 2b_3x + b_5y + \dots \quad \text{and} \quad \frac{\partial P}{\partial y} = b_2 + 2b_4y + b_5x + \dots$$

Now, evaluate them at $(0, 0)$ and set them equal to the derivatives of f :

$$\frac{\partial f}{\partial x}(0, 0) = \frac{\partial P}{\partial x}(0, 0) = b_1 \implies \mathbf{b}_1 = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}(\mathbf{0}, \mathbf{0})$$

$$\frac{\partial f}{\partial y}(0, 0) = \frac{\partial P}{\partial y}(0, 0) = b_2 \implies \mathbf{b}_2 = \frac{\partial \mathbf{f}}{\partial \mathbf{y}}(\mathbf{0}, \mathbf{0})$$

3. **Match the second partial derivatives:** We differentiate our polynomial again:

$$\frac{\partial^2 P}{\partial x^2} = 2b_3 + \dots, \quad \frac{\partial^2 P}{\partial y^2} = 2b_4 + \dots, \quad \frac{\partial^2 P}{\partial y \partial x} = b_5 + \dots$$

Evaluating at $(0, 0)$ gives us the coefficients for the second-order terms:

$$\frac{\partial^2 f}{\partial x^2}(0, 0) = 2b_3 \implies \mathbf{b}_3 = \frac{1}{2} \frac{\partial^2 \mathbf{f}}{\partial \mathbf{x}^2}(\mathbf{0}, \mathbf{0})$$

$$\frac{\partial^2 f}{\partial y^2}(0, 0) = 2b_4 \implies \mathbf{b}_4 = \frac{1}{2} \frac{\partial^2 \mathbf{f}}{\partial \mathbf{y}^2}(\mathbf{0}, \mathbf{0})$$

$$\frac{\partial^2 f}{\partial y \partial x}(0, 0) = b_5 \implies \mathbf{b}_5 = \frac{\partial^2 \mathbf{f}}{\partial \mathbf{y} \partial \mathbf{x}}(\mathbf{0}, \mathbf{0})$$

Substituting these coefficients back gives the Maclaurin series for two variables up to the second order:

$$f(x, y) \approx P(x, y) = f(0, 0) + \frac{\partial f}{\partial x} \Big|_{x=0, y=0} x + \frac{\partial f}{\partial y} \Big|_{x=0, y=0} y + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} \Big|_{x=0, y=0} x^2 + \frac{1}{2} \frac{\partial^2 f}{\partial y^2} \Big|_{x=0, y=0} y^2 + \frac{\partial^2 f}{\partial y \partial x} \Big|_{x=0, y=0} xy$$

(where all derivatives are evaluated at $(0, 0)$).

The Gradient, the Hessian, and the General Taylor Series

The expanded formula above can be written more elegantly using vector and matrix notation. This introduces two important objects:

- **The Gradient (∇f):** A vector containing all the first-order partial derivatives.

$$\nabla f(\vec{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_0} \\ \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

- **The Hessian ($\nabla^2 f$):** A matrix containing all the second-order partial derivatives.

$$\nabla^2 f(\vec{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_0^2} & \frac{\partial^2 f}{\partial x_0 \partial x_1} & \dots \\ \frac{\partial^2 f}{\partial x_1 \partial x_0} & \frac{\partial^2 f}{\partial x_1^2} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Letting $\vec{x} = [x, y]^\top$, we can rewrite the Maclaurin series as:

$$f(\vec{x}) \approx P(\vec{x}) = f(\vec{0}) + \nabla f(\vec{0})^\top \vec{x} + \frac{1}{2} \vec{x}^\top \nabla^2 f(\vec{0}) \vec{x}$$

This is a much more compact representation.

Finally, we don't have to approximate our function around the origin. We can choose any arbitrary point, $\vec{a} = [a, b]^\top$, to be the center of our approximation. This gives rise to the **general Taylor series for multiple variables**, where we measure our displacement from $\vec{a}$ using the term $(\vec{x} - \vec{a})$:

$$f(\vec{x}) \approx P(\vec{x}) = f(\vec{a}) + \nabla f(\vec{a})^\top (\vec{x} - \vec{a}) + \frac{1}{2} (\vec{x} - \vec{a})^\top \nabla^2 f(\vec{a}) (\vec{x} - \vec{a}) + \dots$$

The gradient vector, ∇f , points in the direction of the steepest ascent of the function. This property is the key to an optimization algorithm called **gradient descent**, which is fundamental to training machine learning models and which we will cover in detail in two weeks.